

# Monitoring

1\_Разворачиваем кластер виртуальных машин с помощью файла docker-compose\_cluster.yml

```
[yc-user@kafka-node:~]$ sudo docker ps -a
```

| CONTAINER ID | IMAGE                             | COMMAND                    | CREATED        | STATUS        | PORTS                                                                                                                                           |
|--------------|-----------------------------------|----------------------------|----------------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| 24d7e4132556 | clickhouse/clickhouse-server      | "/entrypoint.sh"           | 25 seconds ago | Up 23 seconds | 9009/tcp, 0.0.0.0:9124->8123/tcp, [::]:9124->8123/tcp, 0.0.0.0:8003->9000/tcp, [::]:8003->9000/tcp, 0.0.0.0:9365->9363/tcp, [::]:9365->9363/tcp |
| 1e40b863bb01 | clickhouse/clickhouse-server      | "/entrypoint.sh"           | 25 seconds ago | Up 23 seconds | 9009/tcp, 0.0.0.0:9123->8123/tcp, [::]:9123->8123/tcp, 0.0.0.0:8002->9000/tcp, [::]:8002->9000/tcp, 0.0.0.0:9364->9363/tcp, [::]:9364->9363/tcp |
| c0f0b0c9b540 | clickhouse/clickhouse-server      | "/entrypoint.sh"           | 25 seconds ago | Up 23 seconds | 9009/tcp, 0.0.0.0:9125->8123/tcp, [::]:9125->8123/tcp, 0.0.0.0:8004->9000/tcp, [::]:8004->9000/tcp, 0.0.0.0:9366->9363/tcp, [::]:9366->9363/tcp |
| 97364acd59ff | confluentinc/cp-kafka:7.8.0       | "/etc/confluent/docker..." | 25 seconds ago | Up 23 seconds | 0.0.0.0:9092->9092/tcp, [::]:9092->9092/tcp, 0.0.0.0:9999->9999/tcp, [::]:9999->9999/tcp, 0.0.0.0:29092->29092/tcp, [::]:29092->29092/tcp       |
| d2d812ead29f | clickhouse/clickhouse-server      | "/entrypoint.sh"           | 25 seconds ago | Up 23 seconds | 9009/tcp, 0.0.0.0:9126->8123/tcp, [::]:9126->8123/tcp, 0.0.0.0:8005->9000/tcp, [::]:8005->9000/tcp, 0.0.0.0:9367->9363/tcp, [::]:9367->9363/tcp |
| c538a003e54a | grafana/grafana-enterprise:latest | "/run.sh"                  | 25 seconds ago | Up 23 seconds | 0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp                                                                                                     |
| c063a0305560 | zookeeper:latest                  | "/docker-entrypoint..."    | 25 seconds ago | Up 23 seconds | 0.0.0.0:2888->2888/tcp, [::]:2888->2888/tcp, 0.0.0.0:3888->3888/tcp, [::]:3888->3888/tcp, 0.0.0.0:2182->2181/tcp, [::]:2182->2181/tcp           |
| 8dc33e684b10 | prom/prometheus:latest            | "/bin/prometheus --c..."   | 25 seconds ago | Up 23 seconds | 0.0.0.0:9090->9090/tcp, [::]:9090->9090/tcp                                                                                                     |

За prometheus отвечает кусок:

```
# =====
# Prometheus
# =====

prometheus:
  image: prom/prometheus:latest
  container_name: prom
  hostname: prometheus
  restart: unless-stopped
  ports:
    - 9090:9090
  volumes:
    - ./config/prometheus/prometheus.yml:/etc/prometheus/prometheus.yml
  networks:
    - otus_ch_net
  environment:
    - TZ= "Europe/Moscow"
  command:
    - --config.file=/etc/prometheus/prometheus.yml
```

С помощью монтирования в docker-compose файле распространяем конфигурацию prometheus на каждую ноду clickhouse

```
- ./config/prometheus.xml:/etc/clickhouse-server/config.d/prometheus.xml
```

## Сам xml

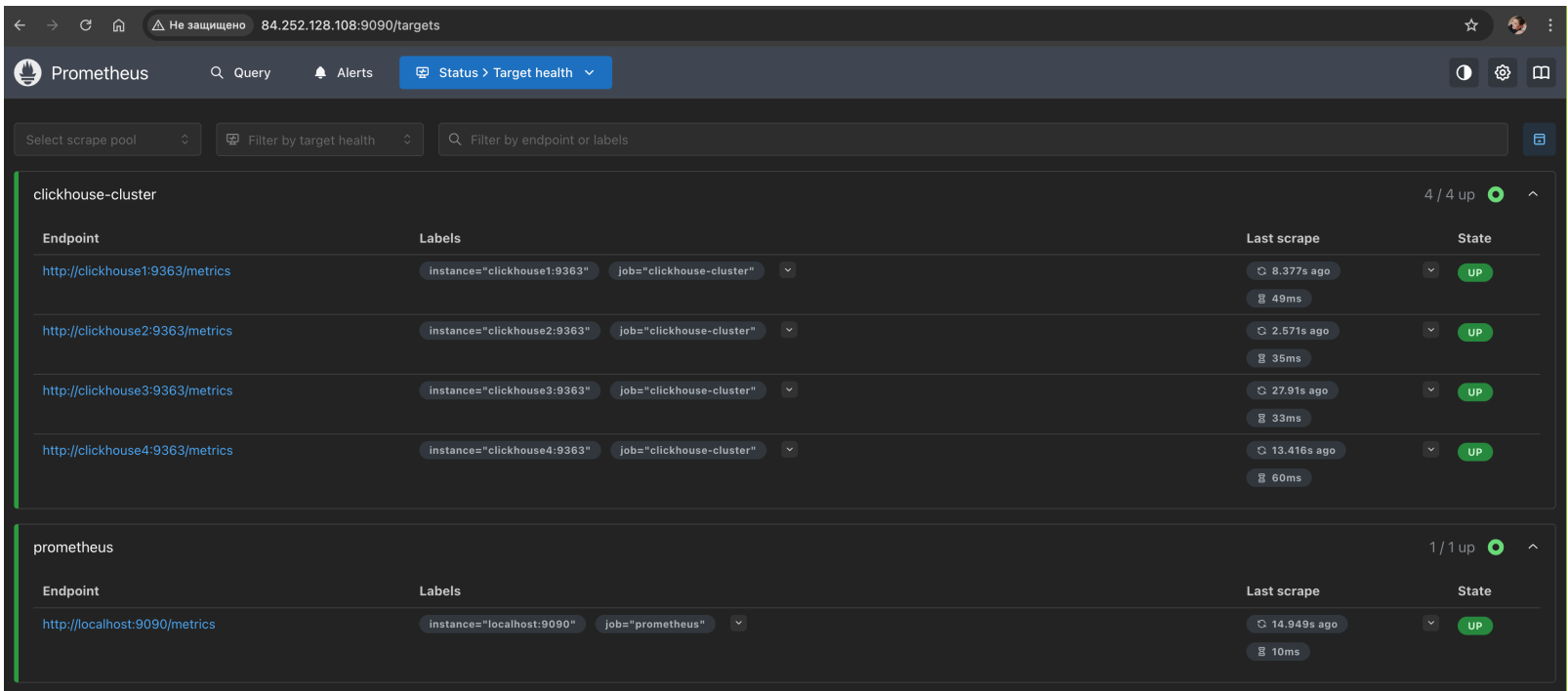
```
<clickhouse>
  <prometheus>
    <endpoint>/metrics</endpoint>
    <port>9363</port>
    <metrics>true</metrics>
    <events>true</events>
    <asynchronous_metrics>true</asynchronous_metrics>
    <errors>true</errors>
  </prometheus>
</clickhouse>
```

Так же создаем и монтируем prometheus.yml

```
global:
  scrape_interval: 30s

scrape_configs:
  - job_name: "prometheus"
    static_configs:
      - targets: ["localhost:9090"]
  - job_name: "clickhouse-cluster"
    static_configs:
      - targets:
        - "clickhouse1:9363"
        - "clickhouse2:9363"
        - "clickhouse3:9363"
        - "clickhouse4:9363"
    metrics_path: "/metrics"
```

## Подключаемся к Prometheus

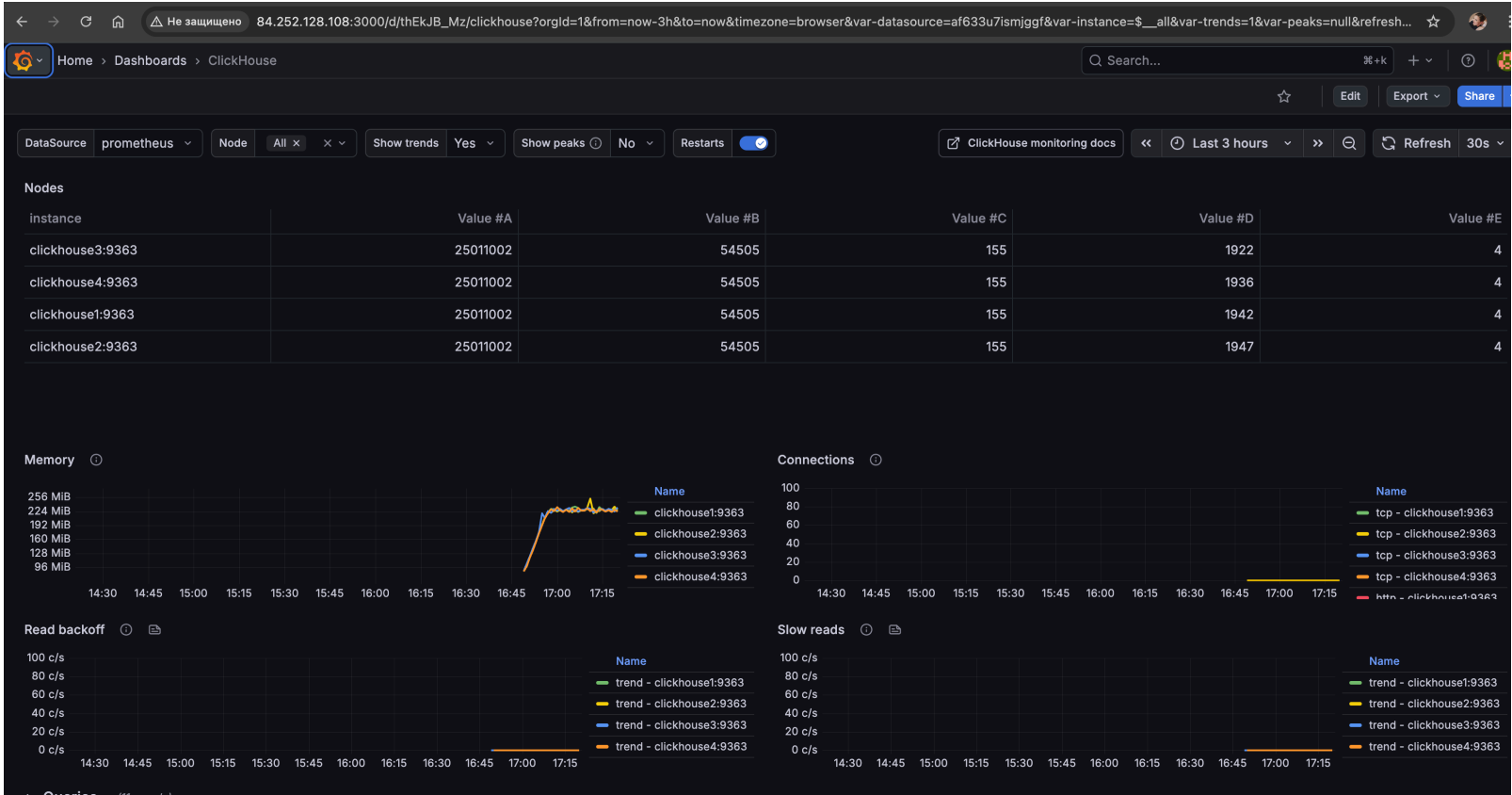


Создаем контейнер с Grafana !!!Важно что пароль должен быть сложным

В grafana UI добавляем источник наш prometheus



И потом создаем дашборд



```
# =====  
# Grafana  
# =====  
grafana:  
  image: grafana/grafana-enterprise:latest  
  container_name: grafana  
  hostname: grafana  
  restart: unless-stopped  
  ports:  
  | - 3000:3000  
  volumes:  
  | - ./config/grafana/provisioning:/etc/grafana/provisioning  
  | - ./config/grafana/dashboards:/etc/grafana/dashboards  
  environment:  
  | - GF_SECURITY_ADMIN_USER=admin  
  | - GF_SECURITY_ADMIN_PASSWORD=****  
  | - TZ="Europe/Moscow"  
  networks:  
  | - otus_ch_net  
  depends_on:  
  | - prometheus
```